CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

Third Semester of B.Tech. (CL) Examination

November 2010

CL 203 Fluid Mechanics

Date: 29.11.2010, Monday **Time: 10:30 a.m. To 01:30 p.m.** **Maximum Marks: 70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define: Specific weight and Specific volume. [2]

(b) How fluid is classified? [3]

(c) Explain with neat sketch the relation between atmospheric pressure, absolute pressure and gauge pressure. [2]

Q - 2 (a) Define: Total Pressure, Center of Pressure and Meta-Centric Height. [3]

(b) A rectangular pontoon is 5.5 m long, 2.5 m wide and 1.2 m high. The depth of immersion of the pontoon is 0.8 m in sea water. If the centre of gravity is 0.6 m above the bottom of the pontoon, determine the meta-centric height. The density for sea water is $1025x10^3$ kg/m^3. [4]

(c) Discuss linear momentum equation. A horizontal venturimeter with inlet and throat diameter 40 cm and 20 cm respectively is used to measure the flow of water. The readings of differential manometer connected to the inlet and throat is 15 cm of mercury. Determine the discharge. Take $C_d = 0.98$ [7]

OR

Q – 2 (a) State hydrostatic law and derive equation for the same. [7]

(b) Discuss momentum correction factor. Also discuss application of momentum equation. [4]

(c) Find the value of oil-mercury differential manometer having $C_d = 0.98$ if discharge of oil flowing through the horizontal venturimeter is 40 litres per second. The venturimeter has inlet and throat diameters 20 cm and 10 cm respectively. Take specific gravity of oil = 0.9. [3]

Q - 3 (a) Discuss and derive continuity equation and energy equation for compressible flow. [6]

(b) An oil of viscosity 0.025 N-s/m^2 flowing between two stationary parallel plates [4]

1.5m wide maintained 15 mm apart. The velocity midway between the plates is 3m/sec. Calculate (a) the pressure gradient along flow (b) the average velocity and (c) discharge for an oil.

(c) The velocity distribution in the boundary layer is given by $\frac{u}{U}=\frac{3}{2}\frac{y}{\delta}-\frac{1}{2}\frac{y^2}{\delta^2}$, where, δ is boundary layer thickness. Calculate (i) the ratio of displacement thickness to boundary layer thickness (δ^*/δ) and (ii) the ratio of momentum thickness to boundary layer thickness (θ/δ). [4]

OR

Q - 3 (a) Find the Mach number, when an aeroplane is flying at 900 km/hr through still air having a pressure of 5 N/cm^2 and temperature -2°C. Wind velocity may be taken as zero. Calculate the pressure and temperature of air at stagnation point on the nose of the plane. Take, k = 1.4 and R = 287 J/kg K. [6]

(b) Derive Darcy–Weisbach's equation of head loss, $h_f=\frac{fLV^2}{2gd}$. [8]

SECTION – II

Q - 4 (a) Derive the equation of forced vortex flow. [4]

(b) An open circular tank of 30 cm diameter and 120 cm long contains water up to a height of 50 cm. The tank is rotated about its vertical axis at 250 r.p.m. Find the depth of parabola formed at the free surface of water. [3]

Q - 5 (a) For a two-dimensional potential flow, the velocity potential is given by Ø = x(2y-1). Determine the velocity at the point P (3, 4). Determine also the value of stream function Ψ at the point P. [5]

(b) Discuss and derive equation for loss of head due to sudden-enlargement in pipe flow. [5]

(c) Water discharges at the rate of 100 litres/sec through a 140 mm diameter vertical sharp-edged orifice placed under a constant head of 12 m. A point, on the jet, measured from the vena-contracta of the jet has co-ordinates 4.3 m horizontal and 0.60 meters vertical. Find the co-efficient C_v, C_c and C_d of the orifice. [4]

OR

Q – 5 (a) Discuss Lagrangian method of motion of fluid particles [5]

(b) The difference in water surface levels in two tanks is 10 m, which are connected by three pipes in series of lengths 250 m, 150 m, 200 m having diameter 300 mm, 190 mm and 350 mm respectively. Determine the rates of flow of water if co-efficient of [5] friction are 0.004, 0.0042 and 0.0038 respectively, considering minor losses also.

(c) Discuss experimental determination of hydraulic coefficient in detail. [4]

Q – 6 (a) Discuss the derivation of discharge through a triangular notch. [5]

(b) A pipe of diameter of 2.0 m is required to transport an oil of specific gravity 0.90 and viscosity 4×10^{-2} poise at the rate of 4000 litres/sec. Tests were conducted on a 20 cm diameter pipe using water at 20^{o}C. Find the velocity and rate of flow in the model. Viscosity of water at 20^{o}C = 0.01 poise. [5]

(c) Classify the hydraulic model. Also discuss scale ratios for distorted models. [4]

OR

Q – 6 (a) Water flows over a rectangular weir 1 m wide at a depth of 140 mm and afterwards passes through a triangular right angled weir. Taking C_d for the rectangular and triangular weir as 0.64 and 0.60 respectively, find the depth over the triangular weir. Take $\theta = 90^{o}$ for triangular right angled weir. [5]

(b) Discuss Rayleigh's & Buckingham's 'π' theorem methods of dimensional analysis in brief. [5]

(c) Find the expression for the power 'P', developed by a pump when 'P' depends upon the head 'H', the discharge 'Q' and specific weight 'w' of the fluid. [4]

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